Event-Driven Risk Intelligence & Portfolio Stress Testing

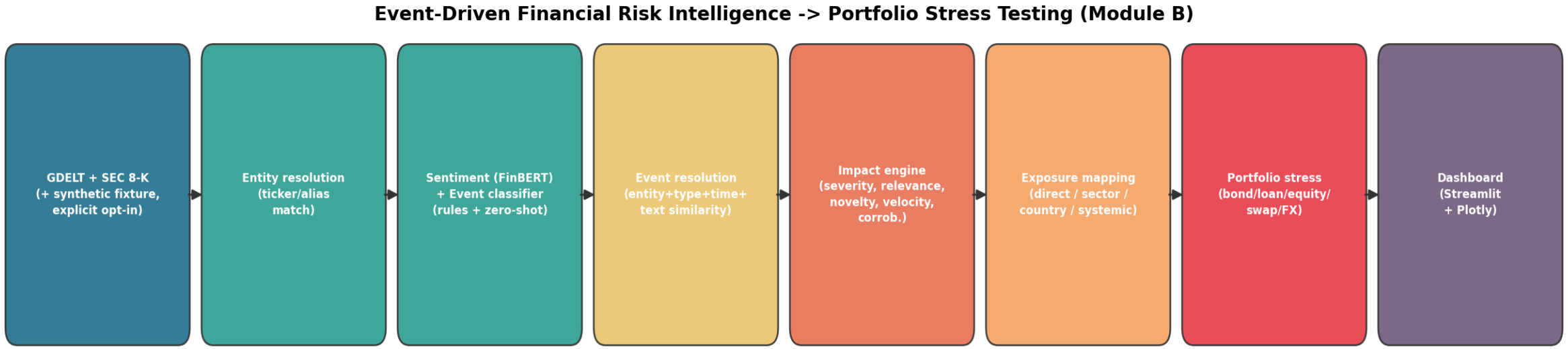
S&P Global & CRISIL Campus Hackathon 2026

From unstructured news & filings to explainable risk signals to synthetic portfolio stress scenarios

Problem & Approach

- Risk teams drown in unstructured news/filings; manual triage is slow and inconsistent.
- Goal: convert raw text into explainable, traceable risk signals - then show their portfolio consequence.
- Core chain: ingestion -> entity resolution -> sentiment + event classification -> event resolution -> impact score -> exposure mapping -> scenario -> asset stress -> portfolio loss.
- Design principles: sentiment (polarity) kept strictly separate from impact (severity); portfolio-agnostic core engine; deterministic replay mode so the demo never depends on a live API; every heuristic documented as a heuristic, never presented as calibrated.

Architecture



Signal contract (JSONL) connects every stage; FastAPI serves it read-only; replay mode makes the whole chain deterministic/offline.

Every stage reuses one shared Signal contract (JSONL); FastAPI exposes it read-only; the synthetic demo fixture is a separate, explicitly-opted-in data path.

Implementation Highlights

- Ingestion: GDELT news + SEC 8-K filings, normalized to one UnifiedDocument schema; deterministic replay data for offline demo.
- NLP: FinBERT sentiment (chunked for long filings) + hybrid rule/zero-shot event classifier across the 12-class taxonomy.
- Event resolution: conservative deterministic clustering (shared entity + same event type + time proximity + title similarity) - never merges on company name alone.
- Impact engine: documented 1-10 heuristic (severity 30% / relevance 25% / novelty 15% / velocity 15% / corroboration 15%), explicitly labeled uncalibrated.
- Module B: synthetic 51-asset portfolio, 8-scenario library, exposure targeting (direct issuer > sector > country > systemic), bond/loan/equity/swap/FX valuation, Streamlit + Plotly dashboard.

Results (Measured)

- 187 automated tests passing in the current build - unit, integration, and full-pipeline tests (entity resolution, sentiment scoring logic, event classification, event resolution, impact math, portfolio valuation mechanics).
- Valuation mechanics (bond duration/convexity, loan expected-credit-loss, equity beta, swap DV01, FX delta) verified against hand-computed examples.
- SYNTHETIC demonstration only (fictional company 'ZZDEMO', fictional news fixture, synthetic portfolio): 5 simulated wire reports -> 1 resolved event -> impact score 9.46/10 -> REGULATORY_LEGAL scenario -> ~$12.5M / 1.7% stylized portfolio loss, correctly concentrated on the directly-exposed issuer.
- NOT measured / not claimed: no real-world predictive accuracy, no backtested market-reaction benchmark, no calibration study - see Limitations.

Domain Impact

- Turns scattered news/filings into one explainable, auditable signal trail: every number traces back to its source document, matched entity, and the rule/model that produced it.
- Separates WHAT happened (event type), HOW SEVERE it looks (impact score), and WHO it actually hits (exposure targeting) - avoiding black-box 'just trust the score' outputs.
- Portfolio-agnostic core risk engine means the same signal can drive different books; Module B shows one concrete, inspectable consumption pattern (stress testing) rather than a vague promise.
- Deterministic replay mode means the full chain - including a guaranteed high-impact scenario - can be demonstrated reliably without depending on live news APIs or model availability during judging.

Limitations & Next Steps

- Impact score, event-severity lookup, exposure-targeting weights, and scenario shock magnitudes are documented, stylized ASSUMPTIONS - not calibrated, not Basel-compliant, not predictive.
- Real FinBERT/zero-shot model inference and long-running server processes (uvicorn, streamlit run) were not executable in this project's network-restricted build environment - verified instead via dependency-injected stub inference and in-process test harnesses; needs one real run on a normal machine before relying on it live.
- Event resolution uses lightweight deterministic text similarity (no embeddings) and can miss some true duplicates/variants by design choice (precision over recall).
- Not built (by design, this phase): Module A tactical rebalancing, VaR, reverse stress testing, advanced derivatives pricing.
- Next: real-model verification run, expand evaluation set (labeled event-classification sample, baseline comparison), broaden scenario/portfolio coverage.